20 The functions f and g are such that

$f(x) = cx + d$ $\quad g(x) = 2x + 5$ where c and d are constants.

$f(4) = 13$ and $g^{-1}(15) = f(2)$

Find the value of c and the value of d.

$4c + d = 13$

1 mark
Use f(4) = 13.

$g^{-1}(15) = (15 - 5) \div 2 = 5$

1 mark
Undo g: −5 then ÷2.

$2c + d = 5$

1 mark
Use f(2) = 5.

Subtract: $2c = 8$, so $c = 4$

$2(4) + d = 5$, so $d = -3$

1 mark
Subtract equations to eliminate d.

1 mark
Substitute c = 4 to find d.

$c =$ 4

$d =$ −3

(Total for Question 20 is 5 marks)

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